TECHNICAL DOCUMENTATION

1. Introduction

Retrieval-Augmented Generation (RAG) combines document retrieval with language models to generate accurate and context-aware responses.

---

2. System Architecture Explanation

The system loads a PDF, splits it into chunks, converts them into embeddings, and stores them in a vector database.
When a query is asked, relevant chunks are retrieved and passed to the LLM for answer generation.

---

3. Design Decisions

- Chunk size = 500 → balance between context and performance
- Overlap = 50 → maintain continuity
- Top-k retrieval = 3 → relevant context selection

---

4. Workflow Explanation

LangGraph manages execution flow using nodes and state transitions.

---

5. Conditional Logic

The system evaluates response quality and decides whether to:

- Return answer
- OR escalate

---

6. HITL Implementation

Human intervention is triggered when:

- Response is unclear
- Context is missing

Benefits:

- Improves reliability
- Reduces incorrect answers

---

7. Challenges & Trade-offs

- GPT-2 gives weaker responses
- Trade-off between speed and accuracy
- Smaller chunks improve speed but reduce context

---

8. Testing Strategy

Sample queries:

- What is refund policy?
- How to contact support?

---

9. Future Enhancements

- Use better LLM (GPT-4)
- Multi-document support
- UI integration